

Nonblocking Multiplexed Digits

Make ghost-prevention order and missed service visible

E0 COPIED POLICY REPLAY — ZERO DISPLAY HARDWARE

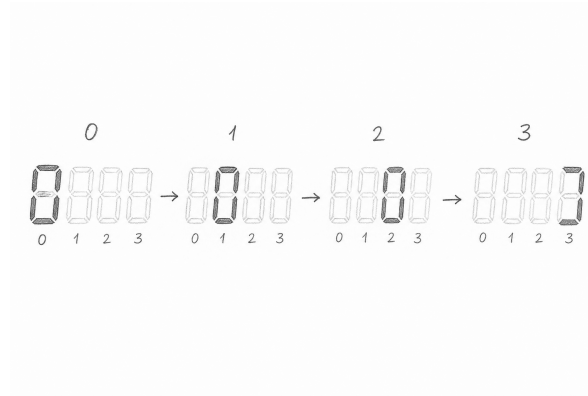
Lesson 058 — component
Prerequisite Lesson 010 glyph vocabulary
Capacity four logical cells

Time 120–180 minutes
Board compile-only Mega replay
Observation named memory result cells

Responsibility. `MultiplexedDigitPolicy` formats one copied value into four logical glyph cells and emits one bounded, ordered refresh transaction when supplied time says service is due. It owns no pin, timer, interrupt, digit driver, display, resistor, supply, or powered circuit.

1 Begin with the illusion

A four-digit display can appear steady even though a controller selects only one digit at a time. The illusion depends on repeated service and on changing the shared segment lines only while every digit select is inactive. Lesson 058 models that ordering as copied intent. It does not claim that any LED lit.



Alternative text: a rough pencil flipbook shows exactly one selected digit moving through zero-based phases 0, 1, 2, and 3 from left to right.

The drawing is a timing mnemonic, not a breadboard layout or formal electrical schematic. At E0, the observation place is the canonical sketch's copied transaction cells.

2 Keep formatting exact

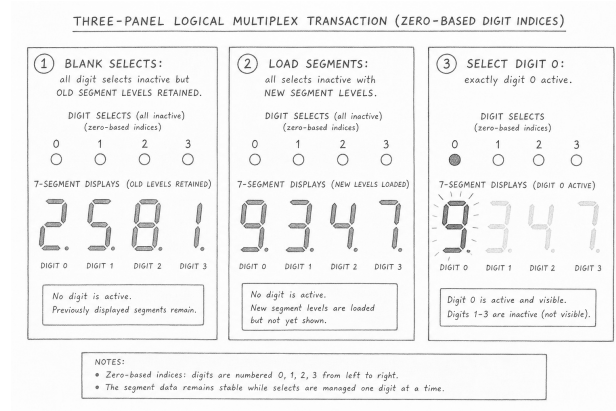
The display has four cells numbered left to right from zero through three. Each cell carries the canonical Lesson 010 segments *a, b, c, d, e, f, g, dp*. Values from 0 through 9,999 render as decimal digits. Larger values render four dashes, clear every decimal point, and report overflow; they never wrap or truncate.

Leading zeros may be shown or blanked, but the value zero always retains one zero. Decimal-mask bit zero names the rightmost cell. Segment polarity and digit-select polarity are separate configuration values. A common-anode or common-cathode label does not reveal the topology of a future digit driver.

Input	Leading-zero policy	Decimal mask	Predict the four cells
0	blank	0x00	blank, blank, blank, zero
42	blank	0x02	blank, blank, four, two with decimal
42	show	0x02	zero, zero, four, two with decimal
9999	either	0x0A	four digits with requested decimal points
10000	either	any	four dashes; no decimal points

3 Prove the three-stage transaction

Each due `refresh(now)` call emits exactly one transaction. First, every digit select is inactive while the prior segment levels remain. Second, every select remains inactive while the next segment levels appear. Third, exactly one digit select becomes active. The order is part of the public evidence because a final mask alone would conceal a ghost-producing update.



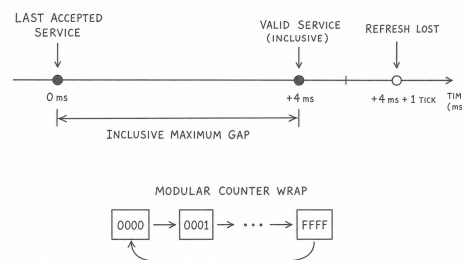
Alternative text: three hand-drawn frames labeled blank, segments, and select show all digit selects inactive, then the next segment mask, then exactly one selected digit. Pencil arrows enforce the order.

Stage	Predict	Inspect in copied evidence
blank	all selects inactive; old segment levels retained	stages[0] select and segment masks
segments	all selects inactive; next segment levels applied	stages[1] select and segment masks
select	one and only one digit active	stages[2] selected index and masks

4 Service one phase, never catch up

The default service period is 1 ms. Calls before the period emit nothing. A due call advances exactly one digit and reanchors to the supplied `now`. It never loops to replay missed phases. Initialization has one explicit exception: its first refresh at the initialization timestamp emits digit zero's complete transaction.

The maximum accepted gap is 4 ms. Equality remains valid; the next tick latches **RefreshLost**, publishes blank intent, and requires explicit reset. Equal timestamps are idempotent after first service. Backward time and an exact unsigned half-range difference reject atomically.



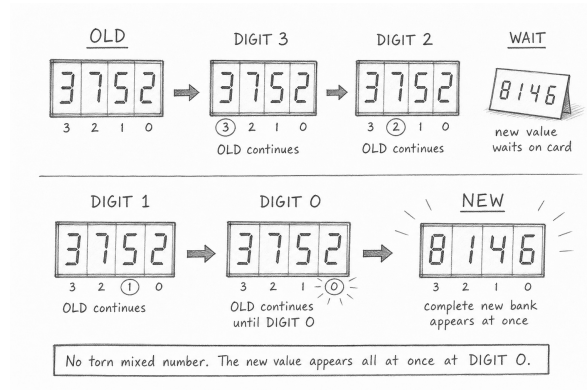
Alternative text: a pencil timeline spans the full interval from the last accepted service to the inclusive four-millisecond boundary; the next tick is separately labeled refresh lost, with modular wrap below.

Supplied time	Prior accepted time	Prediction	Reason
1000	initialized at 1000	digit zero transaction	first-service exception
1000	accepted at 1000	no transaction	equal-time idempotence
1999	accepted at 1000	no transaction	one tick early
2000	accepted at 1000	one transaction	service period reached
6000	accepted at 2000	one transaction	inclusive maximum gap
6001	accepted at 2000	blank, RefreshLost	first invalid gap

5 Publish a frame without tearing

Formatting uses `preview()`, `canCommit()`, and `commit()`. A preview is bound to its owner, lifecycle, configuration, and base generation. Foreign, stale, altered, or consumed previews cannot commit.

A successful commit changes the desired glyph bank, not the bank currently being scanned. The desired bank becomes active only at the next digit-zero boundary. Each bank retains a local generation and the caller-supplied source snapshot sequence.



Alternative text: a hand-drawn current-frame card continues through digits two and three while a desired-frame card waits behind a gate; at digit zero, the complete desired card replaces the complete current card.

Construction is inert. `reset(now)` preserves immutable configuration, invalidates previews, clears the frame, and returns to the initialized blank and digit-zero-armed state. `shutdown()` invalidates previews and is idempotent; service remains inert until later initialization. The policy retains no caller pointer, reads no clock, allocates nothing, recurses nowhere, invokes no callback, and performs bounded work.

6 Read the canonical replay

The canonical compile-only Mega sketch follows acquire, configure, start and then observe, decide, actuate. Its “actuation” is assignment to named volatile result cells:

```
MultiplexedDigitPreview candidate;
const Status previewStatus =
    display.preview(value, showLeadingZeros,
                   decimalMask, sourceSequence,
                   candidate);
const Status commitStatus = previewStatus.ok()
    ? display.commit(candidate) : previewStatus;
const Result<MultiplexedDigitTransaction> result =
    display.refresh(now);
```

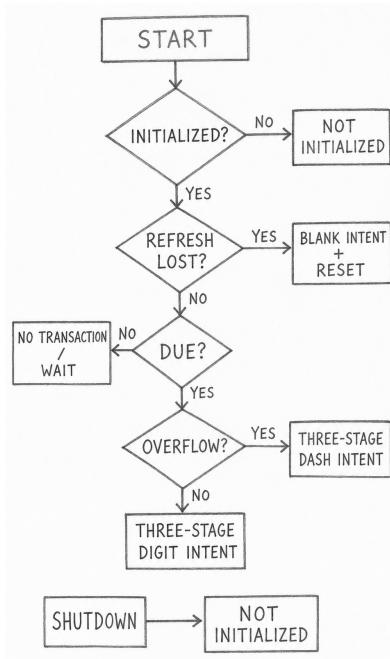
The exact source is downloadable as text. Do not credit Serial output. A host test may compare every field of a copied transaction; the sketch offers the same logical evidence through inspectable memory cells.

7 Run predict, observe, interpret

1. **Initialize.** Predict blank intent, digit zero armed, and no hardware claims. Observe lifecycle, blank, and generation cells. Interpret success as E0 policy initialization only.
2. **First refresh.** Predict blank/segments/select order at the initialization timestamp. Observe all three copied stage cells. Interpret their order; do not infer a waveform.
3. **Format.** Predict the complete bank for zero, 42 with and without leading zeros, 9,999, and overflow. Observe desired glyphs, decimal mask, overflow, source sequence, and generation.
4. **Swap.** Commit while digit three is active. Predict that the old bank finishes and the new bank appears only at digit zero. Observe desired and active generations at each service.
5. **Timing.** Exercise early, due, equal, wraparound, exact 4 ms, and one-tick-late calls. Observe disposition, phase, and blank intent.
6. **Recover.** Predict that refresh loss remains latched until reset. Observe inert service before reset, then blank and armed state.
7. **Shutdown.** Predict no later transaction. Observe lifecycle and copied blank intent. Interpret no physical blanking claim.

Case	time	active generation	digit	result / interpretation
first service	_____	_____	_____	_____
early	_____	_____	_____	_____
digit-zero swap	_____	_____	_____	_____
exact gap limit	_____	_____	_____	_____
one tick late	_____	_____	_____	_____

8 Diagnose from the first violated contract



Alternative text: a sparse graphite E0 decision tree checks initialization, refresh loss, whether service is due, and overflow. Leaves identify no transaction, blank intent, dash intent, or digit intent.

Symptom	Inspect first	Do not conclude
initialization rejects	period, maximum gap, polarity, configuration ID	display is electrically faulty
no transaction	lifecycle, first-service flag, elapsed period	the processor or display stopped
wrong-looking copied mask	glyph, decimal, segment polarity	future driver topology is known
old value remains	desired versus active generation and next digit	commit tore the frame
blank intent is latched	refresh fault and last accepted service time	physical LEDs are dark
changed preview rejects	owner, lifecycle, configuration, base generation	part of the new bank committed

9 Separate acquisition from safe-state evidence

Acquisition prediction. A valid fixed configuration initializes and owns no resources. **Observe** initialization status, lifecycle, configuration revision, blank snapshot, and an unchanged resource-registry fixture. **Interpret** that evidence as bounded software setup, not pin ownership or physical execution.

Safe-state prediction. Construction, failed initialization, reset, refresh loss, and shutdown expose copied blank intent and inactive digit selects. **Observe** the complete logical snapshot and transaction fields. **Interpret** them as policy state only. They do not establish high impedance, zero current, physical blanking, or power isolation.

10 Record deterministic and resource proof

Host tests must cover every canonical glyph; leading-zero and decimal combinations; all four segment/select polarity combinations; exact blank/segments/select order; digit sequence; digit-zero frame swap; early, due, equal, late, rollover, regression, and half-range time; lifecycle and preview collisions; refresh-loss recovery; shutdown; and byte-identical replay.

The exact no-LTO AVR probe records flash, static SRAM, reusable object size, conservative synchronous stack, residual SRAM, compiler/core/flag fingerprints, and the maximum live fixture. The initial target/hard ceilings are 12/16 KiB flash, 768/1,024 bytes static SRAM, 320/448 bytes synchronous stack, and 192/256 bytes for the reusable object.

Evidence	Measured	Target / hard	Disposition
flash	6,044 B	12 / 16 KiB	pass
static SRAM	759 B	768 / 1,024 B	pass
reusable object	59 B	192 / 256 B	pass
synchronous stack	187 B	320 / 448 B	pass

These are compiler-derived E0 measurements, not runtime stack-watermark, electrical, optical, or physical evidence. Re-measure after any change to the source, sketch, fixture, compiler, core, flags, or composition. A target miss requires a stale-failing reviewed marker; a hard or residual miss blocks promotion.

11 Exercises

1. Draw the four-cell result for 7 with leading zeros off and decimal masks 0x01, 0x08, and 0x0F. Then compare every glyph and decimal bit with a preview.
2. For active-high segments and active-low digit selects, calculate all three masks for digit two showing 8. Repeat with both polarities inverted. Keep logical and electrical levels distinct.
3. Commit a new bank while digit one is active. Write the next four refresh digit indices and identify exactly where the generation may change.
4. Construct a supplied-time trace that crosses unsigned wrap while every duration remains unambiguous. Add one exact-half-range call and predict which state may change.
5. Explain why replaying four missed digits in one call would weaken both bounded execution and evidence about actual service cadence.

12 Keep the powered fixture open

Lesson 058 is E0, so it deliberately contains no wiring table, breadboard layout, or formal schematic. E1a remains open until the exact four-digit specimen, polarity, pin map, segment network, and digit-driver topology are identified from authoritative evidence.

The E1a record must use one resistor per segment and a rated digit-driver topology rather than sending aggregate common current through a Mega pin. It must calculate and measure peak segment current, multiplex duty, per-pin and per-port current, digit-driver peak and continuous ratings, worst-case lit segments, loaded rail and USB current, refresh waveform, brightness, ghosting, reset, and blanking. It must prove transactional resource acquisition and rollback separately from safe-state voltage/current evidence, and it must include an electrically authoritative formal schematic, all-power-off inspection, stop conditions, and physical power removal.

Open E1a item	Required record before power
specimen identity	exact marking, polarity, datasheet, orientation
segment paths	one resistor per segment; value and current calculation
digit paths	exact rated driver topology and current ratings
resources	pins, ports, timers/interrupts if any, rollback sequence
observability	segment and digit-select test points plus expected waveforms
safe state	startup, reset, fault, shutdown, and power-removed observations
acceptance	person, board, supply, instruments, versions, date, deviations

Until every row is resolved, keep the display unpowered. Software blank intent, reset, shutdown, and destruction are not emergency stops and do not prove that a powered display is dark.

13 Evidence and next step

Compare the canonical sketch's named result cells with deterministic host tests and the accessible HTML reference. Record the frame generation _____, source sequence _____, last accepted phase _____, refresh disposition _____, and safe-state interpretation _____.

Lesson 059 adds bounded MAX7219 register intent while preserving transport prefixes and blank-request evidence. Lesson 060 derives both displays from one immutable stopwatch snapshot and reports disagreement rather than claiming simultaneous photons.